

The goal is to prove convergence of the model in the forme of the following estimates :

$$\mathbb{E}^{\frac{1}{p}} |\Pi_{x\beta r}(x)|^p \lesssim r^{|\beta|}, \quad (1)$$

$$\mathbb{E}^{\frac{1}{p}} |\Pi_{x\beta r}^-(x)|^p \lesssim r^{|\beta|-2}, \quad (2)$$

$$\mathbb{E}^{\frac{1}{p}} |(\Gamma_x^*)_\beta^\gamma|^p \lesssim r^{|\beta|-|\gamma|} \quad \text{for all populated } \gamma. \quad (3)$$

The first two estimates are homogeneity estimates on the germs Π_x and Π_x^- , and together with the last one, we also get a coherence estimate on these two germs.

$$\mathbb{E}^{\frac{1}{p}} |(\Pi_{x+y\beta} - \Pi_{x\beta})_r(x)|^p \lesssim r^\alpha (r + |y|)^{|\beta|-\alpha}, \quad (4)$$

$$\mathbb{E}^{\frac{1}{p}} |\Pi_{\beta r}^-(x)|^p \lesssim r^{3\alpha} (r + |x|)^{|\beta|-2-3\alpha}, \quad \text{provided } \beta \neq 0. \quad (5)$$

This estimate gives us useful insight : the increments $\Pi_{x+y\beta} - \Pi_{x\beta}$ have local regularity α (which is negative) at an arbitrary point, but vanishes at the order $|\beta|$ when y goes to 0. Finally, at infinity, they grow at order $|\beta| - \alpha$.

Proof. Proof of (4) We know that $\Pi_{x+y\beta} - \Pi_{x\beta} = (\Gamma_y^* - id)\Pi_{x\beta} = \sum_\gamma (\Gamma_y^* - id)_\beta^\gamma \Pi_{x\gamma}$. We apply the mollification operator $(\cdot)_r$ and evaluate at x . Using the triangle inequality with respect to the norm $\mathbb{E}^{\frac{1}{p}} |\cdot|^p$, and then Hölder's inequality, we get :

$$\mathbb{E}^{\frac{1}{p}} |(\Pi_{x+y\beta} - \Pi_{x\beta})_r(x)|^p \lesssim \sum_\gamma \mathbb{E}^{\frac{1}{2p}} |(\Gamma_y^* - id)_\beta^\gamma|^{2p} \mathbb{E}^{\frac{1}{2p}} |\Pi_{x\gamma r}(x)|^{2p}. \quad (6)$$

Now appealing to $|\gamma| \geq |\beta| \implies (\Gamma_y^* - id)_\beta^\gamma = 0$ to restrict the sum and get rid of id , and estimates (1) and (3), we get :

$$\mathbb{E}^{\frac{1}{p}} |(\Pi_{x+y\beta} - \Pi_{x\beta})_r(x)|^p \lesssim \sum_{|\gamma| \leq |\beta|} |y|^{|\beta|-|\gamma|} r^{|\gamma|}. \quad (7)$$

We also know that $|\gamma| \geq \alpha$ for all populated γ (see (62) in [BOT25], α is the roughest regularity of the model, it is the regularity of Π_0), so we can restrict the sum to $|\gamma| \geq \alpha$. Factorizing by r^α , and using a binomial expansion, we get (4). $\square$

The proof for Π^- is analog.

We are not trying to apply a reconstruction theorem here, but coherence and homogeneity are the two relevant metrics to measure germs of distributions. In order to prove these estimates, we will first prove similar estimates on $\delta\Pi$ and $\delta\Pi^-$, which are the Malliavin derivatives of Π and Π^- with respect to the noise ξ , or rather on their germs $d\Gamma_x^* \Pi_x$ and $d\Gamma_x^* \Pi_x^-$ (cf section 2.6 from [BOT25] to see why these are the right germs). We will take advantage of the fact that $\delta\Pi$ and $\delta\Pi^-$ are related by a robust relation, and that they are more regular, by a order $\frac{D}{2}$. The shape of the estimate will be slightly different as we are going to take a local L^2 norm in x instead of an L^∞ norm. We will thus try to prove the following estimates :

$$\left(\int_{B_R} dx \mathbb{E}^{\frac{2}{q}} \left| \sum_{|\gamma| \geq 2+s} (d\Gamma_x^*)_\beta^\gamma \Pi_{x\gamma r}^-(x) \right|^q \right)^{\frac{1}{2}} \lesssim r^{\alpha-2+\frac{D}{2}} (r+R)^{|\beta|-\alpha} \quad (\text{homogeneity}) \quad (8)$$

$$\begin{aligned}
& \left(\int_{B_R} dx \mathbb{E}^{\frac{2}{q}} |(\mathrm{d}\Gamma_{x+y}^* \Pi_{x+y}^- - \mathrm{d}\Gamma_x^* \Pi_x^-)_{\beta r}(x+y)|^q \right)^{\frac{1}{2}} \\
& \lesssim r^{2\alpha} (r + |y|)^{\alpha + \frac{D}{2}} (|y| + r + R)^{|\beta| - 2 - 3\alpha} \quad (\text{coherence}) \quad (9)
\end{aligned}$$

and

$$\left(\int_{B_R} dx \mathbb{E}^{\frac{2}{q}} \left| \sum_{|\gamma| \geq 2+s} (\mathrm{d}\Gamma_x^*)_{\beta}^{\gamma} \Pi_{x\gamma r}(x) \right|^q \right)^{\frac{1}{2}} \lesssim r^{\alpha + \frac{D}{2}} (r + R)^{|\beta| - \alpha} \quad (\text{homogeneity}) \quad (10)$$

This will allow us to prove by reconstruction that :

$$\left(\int_{B_R} dx \mathbb{E}^{\frac{2}{q}} |(\delta\Pi^- - \mathrm{d}\Gamma_x^* \Pi_x^-)_{\beta r}(x)|^q \right)^{\frac{1}{2}} \lesssim r^{3\alpha + \frac{D}{2}} (r + R)^{|\beta| - 2 - 3\alpha} \quad (11)$$

And we won't need a coherence estimate for $\mathrm{d}\Gamma_x^* \Pi_x^-$ as we will directly deduce a reconstruction bound by integration of the one we get for $\delta\Pi^-$:

$$\left(\int_{B_R} dx \mathbb{E}^{\frac{2}{q}} |(\delta\Pi - \mathrm{d}\Gamma_x^* \Pi_x)_{\beta r}(x)|^q \right)^{\frac{1}{2}} \lesssim r^{\alpha + \frac{D}{2}} (r + R)^{|\beta| - \alpha} \quad (12)$$

What is crucial here is the exponent $3\alpha + \frac{D}{2}$, in the coherence estimate of $\mathrm{d}\Gamma_x^* \Pi_x^-$ and in the reconstruction bound for $\delta\Pi^-$. It indeed plays the role of γ in the reconstruction theorem presented in [BL23], and is positive by assumption in [BOT25]. Because of this, the reconstruction theorem ensures uniqueness of the reconstruction $\delta\Pi^-$. The same goes for $\delta\Pi$ with the exponent $\alpha + \frac{D}{2}$ which is even more positive as $\alpha < 0$. This cascades down to show that our model is uniquely characterized.

Bibliography

- [BL23] Lucas Broux and David Lee. Besov reconstruction. *Potential Analysis*, 59(4):1875–1912, 12 2023.
- [BOT25] Lucas Broux, Felix Otto, and Markus Tempelmayr. Lecture notes on Malliavin calculus in regularity structures. *Stochastics and Partial Differential Equations: Analysis and Computations*, 09 2025.